

Multinomial logits - model for categorical outcomes

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GLM: A recap

Reminder: What is a GLM?

Regressions aim to describe (a linear) relationship between x and y with one number, β .

- ▶ Assumes a continuous and unbounded variable.
- ▶ When y is neither (e.g. binary), we relied on a latent continuous variable
- ▶ To approximate the latent variable, we calculated the logodds (i.e. we compare)

⇒ *Probability distribution maps unobserved variable to observed outcomes.*

Topic (next weeks): nominal and ordinal variables

GLMs when our outcome variable is categorical

- ▶ categorical (e.g. party, profession, . . .) $\rightarrow$ *multinomial or conditional regression*
- ▶ ordinal (e.g. attitudes towards topics. . .) $\rightarrow$ *ordinal regression*

Dependent variable: nominal

The discrete choice models describe mutually exclusive choices.

- ▶ The choice variable is nominal: we cannot rank it
- ▶ Our *appreciation* of it is continuous. Two sets of models:
 - ▶ Multinomial: Models *chooser* characteristics
 - ▶ Conditional logit: Models *choice* characteristics

Two conceptions of multinomial regression

Two conceptions of multinomial regression

- ▶ **Latent variable approach:** Our utility of each choice.
- ▶ **A series of binomial logits** with the same reference category.

Latent variable approach

Latent variable approach: Imagine m choices modeled as

$$y_m = a_m \times b_m x_i$$

- ▶ $b_m x_i$ reflects the utility of a choice m for the chooser i with x characteristic. $\rightarrow$ systematic term
- ▶ a_m reflects the baseline utility of that choice $\rightarrow$ stochastic term

$\Rightarrow$ *The preferred choice is the one with the highest utility*

Example: Party choice

Example: Party choice

Let's consider party choice among voters

- ▶ ESS survey round (chap 6, Hermansen, 2023)
- ▶ respondents give:
 - ▶ preferred party
 - ▶ attitudes towards immigration

I can rank parties

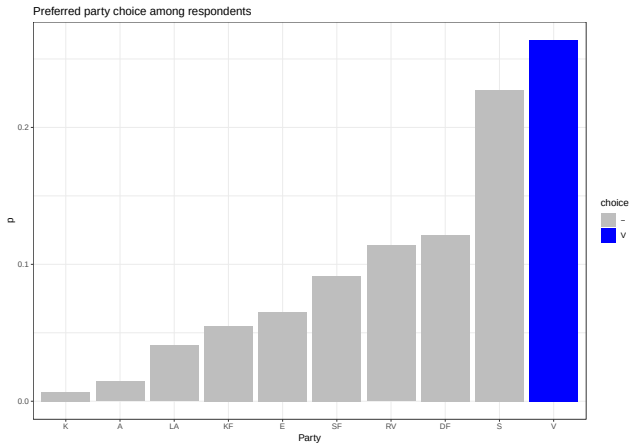
Let's rank the parties according to the respondents' choice

```
tab <-
  df %>%
    #Group by party
    group_by(Party) %>%
    #Number of respondent by party
    reframe(n = n()) %>%
    mutate(
      #Total number of respondents
      N = sum(n),
      #Proportion/probability of group
      p = n/N) %>%
    #Sort just for facility
    arrange(p) %>%
    mutate(
      #Check if it sums up to 1
      cum = cumsum(p),
      #Which is the largest?
      choice = if_else(row.names(.) == which.max(p),
        Party, "-"))
```

I can rank parties

```
## # A tibble: 10 x 6
##   Party      n      N      p      cum choice
##   <chr> <int> <int> <dbl> <dbl> <chr>
## 1 K         8    1179 0.00679 0.00679 -
## 2 A        17    1179 0.0144 0.0212 -
## 3 LA        48    1179 0.0407 0.0619 -
## 4 KF        65    1179 0.0551 0.117 -
## 5 E         77    1179 0.0653 0.182 -
## 6 SF       108    1179 0.0916 0.274 -
## 7 RV       134    1179 0.114 0.388 -
## 8 DF       143    1179 0.121 0.509 -
## 9 S       268    1179 0.227 0.736 -
## 10 V      311    1179 0.264 1      V
```

I can rank parties (figure)



- ▶ the most frequent party choice is the most probable outcome

Theoretical link to political science

The assumption is that choosers are rational, and choose a category (m_j) whenever its utility exceeds the alternative (m_d).

$$U(m_j) > U(m_d)$$

⇒ This is also how we estimate it; through comparisons

A series of binomial logits

A series of binomial logits with the *same* reference category.

- ▶ Data consists of many groups, but I only compare two groups → data/variation intensive model choice.
- ▶ Categories/choice are mutually exclusive → Different β for each choice

⇒ *All choices are given a probability and they sum up to one.*

Example: ESS survey round

Let's do an intercept-only model

Logit transformation:

$$\text{logit}(p_m) = \log\left(\frac{p_m}{p_d}\right)$$

```
tab <-  
  df %>%  
    #Group by party  
    group_by(Party) %>%  
    #Number of respondent by party  
    reframe(n = n()) %>%  
    mutate(  
      #Total number of respondents  
      N = sum(n),  
      #Proportion/probability of group  
      p = n/N,  
      #Pick Social democrats as reference category  
      p_ref = p[Party == "S"],  
      #Odds  
      odds = p/p_ref,  
      #Logodds  
      logodds = log(odds))
```


Example: ESS survey

- ▶ intercept-only model
- ▶ ... where the reference-level (S) is effectively left out

```
## # A tibble: 10 x 7
##   Party      n      N      p p_ref odds logodds
##   <chr> <int> <int>   <dbl> <dbl>   <dbl>   <dbl>
## 1 A      17   1179 0.0144 0.227 0.0634 -2.76
## 2 DF    143   1179 0.121 0.227 0.534 -0.628
## 3 E      77   1179 0.0653 0.227 0.287 -1.25
## 4 K       8   1179 0.00679 0.227 0.0299 -3.51
## 5 KF     65   1179 0.0551 0.227 0.243 -1.42
## 6 LA     48   1179 0.0407 0.227 0.179 -1.72
## 7 RV    134   1179 0.114 0.227 0.5 -0.693
## 8 S    268   1179 0.227 0.227 1 0
## 9 SF    108   1179 0.0916 0.227 0.403 -0.909
## 10 V    311   1179 0.264 0.227 1.16 0.149
```

In R: set a reference level

- ▶ We set a reference level p_d : That's the leave-one-out trick.

```
df <-  
df %>%  
  #I use the Social democrats  
  mutate(Party = relevel(as.factor(Party), ref = "S"))
```

- ▶ Estimate the model

```
library(nnet)  
mod.cat <- multinom(Party ~  
  1,  
  df)
```

```
## # weights:  20 (9 variable)  
## initial value 2714.747825  
## iter  10 value 2332.511892  
## final value 2326.831829  
## converged
```

Results table

The result is a series of equations, one for each party

Table 1:

	<i>Dependent variable:</i>						
	A (1)	DF (2)	E (3)	K (4)	KF (5)	LA (6)	RV (7)
Constant	-2.76*** (0.25)	-0.63*** (0.10)	-1.25*** (0.13)	-3.51*** (0.36)	-1.42*** (0.14)	-1.72*** (0.16)	-0.69*** (0.11)
Akaike Inf. Crit.	4,671.66	4,671.66	4,671.66	4,671.66	4,671.66	4,671.66	4,671.66

Note:

With predictors

Let's regress party choice on scepticism towards immigration

```
library(nnet)
mod.cat <- multinom(Party ~
  Skepsis,
  df)
```

```
## # weights: 30 (18 variable)
## initial value 2705.537484
## iter 10 value 2304.290245
## iter 20 value 2246.392642
## final value 2246.301290
## converged
```

Table 2:

	<i>Dependent variable:</i>						
	A (1)	DF (2)	E (3)	K (4)	KF (5)	LA (6)	RV (7)
Skepsis	0.23 (0.15)	0.56*** (0.07)	-0.04 (0.08)	-0.18 (0.24)	0.04 (0.09)	0.07 (0.10)	-0.30*** (0.07)
Constant	-3.89*** (0.85)	-3.69*** (0.40)	-1.04** (0.41)	-2.70** (1.09)	-1.58*** (0.44)	-2.06*** (0.51)	0.62* (0.32)
Akaike Inf. Crit.	4,528.60	4,528.60	4,528.60	4,528.60	4,528.60	4,528.60	4,528.60

Interpretation

Interpretation

All the possibilities of the binomial logit are open, but the backtransformation is a hack

⇒ However, you want to decide which story you want to tell

Different approaches

- ▶ With respect to the reference category
 - ▶ the regression table (logodds): direction and statistical significance
 - ▶ marginal effects (partial back-transformation): relative change
- ▶ Predicted outcomes per category
 - ▶ predicted probability of each category (transformation of latent variable): when one increases, the other decrease
 - ▶ predicted choice (total back-transformation): most probable choice

Marginal effects

The marginal effects are interpreted with reference to the reference level:

- ▶ A one-unit increase in skepticism decreases the probability of voting Alternativet rather than Social democrats with:
 - ▶ $(1 - \exp(0.23) \times 100) = -25\%$

Predicted probabilities

The results can be read as a series of equations, one for each category m

$$Pr(y = m) \sim \log(odds)$$

$$\log(odds) = a_m + b_mx$$

- predictions for each category $\rightarrow$ *separate slopes and intercept*

$$\log(odds) = -3.89 + 0.23x$$

$\Rightarrow$ *The “latent” variable is here represented by the logodds*

Predicted probabilities (cont.)

The manual backtransformation requires more manual work

1. set a scenario (e.g. $x = 5$)
 2. backtransform: divide the odds for the relevant category by the sum of the odds for all categories (incl. the reference) within each scenario
- calculate the logodds by hand for all categories within the scenario, sum over and exponentiate
- ... or use the `predict()` function in R

```
preds <- predict(mod.cat, newdata = data.frame(Skepsis = 5), type = "probs")
preds
```

```
##           S           A           DF           E           K           KF
## 0.238097927 0.015061927 0.096815641 0.067125356 0.006603907 0.058599993
##           LA           RV           SF           V
## 0.043437070 0.100558519 0.093078122 0.280621538
```

⇒ The probability that a respondent with moderate view on immigration votes Alternativet is 2 %

Predicted probabilities using R

Predictions give latent probability of voting for a party, given the scenario.

- ▶ quickly many predictions

```
predict(mod.cat, newdata = data.frame(Skepsis = 0:10), type = "probs") %>% round(., 3)
```

##	S	A	DF	E	K	KF	LA	RV	SF	V
## 1	0.202	0.004	0.005	0.071	0.014	0.041	0.026	0.374	0.132	0.132
## 2	0.221	0.006	0.010	0.075	0.012	0.047	0.030	0.306	0.130	0.163
## 3	0.236	0.008	0.018	0.076	0.011	0.052	0.035	0.243	0.126	0.196
## 4	0.245	0.010	0.033	0.075	0.010	0.056	0.039	0.187	0.118	0.228
## 5	0.246	0.012	0.057	0.072	0.008	0.058	0.042	0.140	0.107	0.258
## 6	0.238	0.015	0.097	0.067	0.007	0.059	0.043	0.101	0.093	0.281
## 7	0.221	0.018	0.157	0.060	0.005	0.056	0.043	0.069	0.078	0.293
## 8	0.195	0.019	0.242	0.050	0.004	0.051	0.041	0.045	0.062	0.290
## 9	0.161	0.020	0.350	0.040	0.003	0.044	0.036	0.028	0.046	0.271
## 10	0.125	0.020	0.474	0.029	0.002	0.036	0.030	0.016	0.032	0.236
## 11	0.090	0.018	0.599	0.020	0.001	0.027	0.023	0.009	0.021	0.192

- ▶ in each scenario the sum of probabilities is one:
 - ▶ when the probability of voting for one party increases, the probability decreases for other parties
 - ▶ lines of effect plot become dependent

Total backtransformation

To predict party choice, I identify the party with the highest probability within each scenario/respondent

- ▶ I let the scenario vary (or I can do in-sample prediction) and predict probabilities

```
preds <- predict(mod.cat, newdata = data.frame(Skepsis = 0:10), type = "probs")
```

- ▶ I identify the most likely outcome for scenario 1

```
## RV
## 8
```

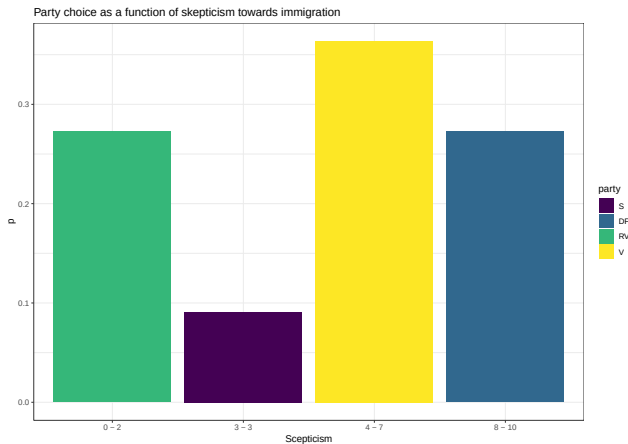
- ▶ R can also do it for me

```
preds <- predict(mod.cat, newdata = data.frame(Skepsis = 0:10), type = "class")
preds
```

```
## [1] RV RV RV S V V V V DF DF DF
## Levels: S A DF E K KF LA RV SF V
```

Total backtransformation (cont.)

Predicted categorical outcomes can also be illustrated in a barplot, with the predictor on the x-axis



Main assumption: IIA

Independence of irrelevant alternatives:

- ▶ there are no choices beyond what is modeled
- ▶ consistency: if we prefer $A > B$ and $B > C$, then also $A > C$

⇒ *The β does not depend on other values of y (other alternatives).*

Testing the main assumption:

The Hausmann-McFadden test: Removes an alternative (supposed to be irrelevant) and check if β changes.

- ▶ Restricted model (a choice is removed) vs. unrestricted model (original)
- ▶ if IIA holds, then unrestricted model has smaller variance.

$\Rightarrow \chi^2$ -test with smaller value indicates IIA holds.

Prediction testing

► Predict outcome

- predicted outcome/choice is the one with the highest probability/utility
- confusion matrix (Proportion of correct predictions: $\frac{\text{sum of diagonal}}{N \text{ observations}}$)

► Probability of all outcomes separately: ROC curve and separation plots

⇒ *as in binomial regression, where you have one category vs. the rest*